

TEJAS NALADALA

Hardware Engineer · AI Builder · Startup Founder

Seattle, WA · (206) 963-2520 · tejas.naladala@gmail.com ·

tejasnaladala.com · linkedin.com/in/tejasnaladala · github.com/tejasnaladala · [Google Scholar](https://scholar.google.com/citations?user=tejasnaladala)

PROFILE

Hardware engineer and AI systems builder shipping \$180K plasma reactors and autonomous research engines before graduating college. Founded **PlasmaX** at 16 (closed \$2M seed at \$8M post-money, sold first production unit to CSIR-CFTRI India, 3 peer-reviewed publications, patent filed). Currently leading **Parchment Labs** (autonomous research discovery engine) and **delphi** (verification pipeline for LLM-generated quantitative strategies). Lavin Entrepreneurship Fellow at the University of Washington.

Research interests: safety-relevant evaluation and interpretability of frontier models; empirical study design for failure modes in multi-agent and RL systems; reproducible research artifacts (pre-registered protocols, open code and data, audit trails).

EDUCATION

University of Washington, Seattle

Sep 2024 – Jun 2028

B.S. Electrical & Computer Engineering and Applied Mathematics (Double Major) · GPA 3.93 / 4.0

- Lavin Entrepreneurship Fellow (four-year selective program)
- Y Combinator AI Startup School 2026 (SF cohort)
- Activities: Research & Computing Club · National Acapella Team

RESEARCH & FOUNDER EXPERIENCE

Parchment Labs · Founder & Lead Researcher

Dec 2025 – Present

Reproducible literature-to-experiment pipeline for ML research · Autonomous research discovery and experimentation engine

- Built a 9-stage autonomous research pipeline: multi-source ingestion (ArXiv, bioRxiv, Semantic Scholar: 50+ categories, 5,000+ papers/day), vector embedding via a local 32B-parameter model with HNSW indexing, novelty/relevance scoring with cosine deduplication, knowledge-graph gap detection, multi-agent adversarial peer review (150 agents, weighted endorsement voting, 85% consensus threshold), experiment design, GPU execution, and LaTeX manuscript generation.
- Connectome Architecture Benchmark (Nature Communications submission). Compared 10 published connectomes (C. elegans, Drosophila, Ciona intestinalis, + 7 others) against degree-preserving, Erdős–Rényi, Watts–Strogatz, and Barabási–Albert nulls on 6 downstream tasks (900 training runs, bootstrap CIs). Degree-preserving nulls recover most of the advantage; residual +7.4% improvement on mushroom-body tasks and a 14.7% wallclock speed-up during training.
- Replayable research pipeline: every run reproducible from commit hash + config. Proprietary multi-agent review layer with 1,100+ self-learning calibration patterns from structured rejection reasoning. Zero-API local inference on a 32B open model at 93%+ gross margins.

delphi · Founder & Lead Engineer

Jan 2026 – Present

Multi-stage verification pipeline for LLM-generated quantitative strategies

- Medallion-architecture data plane (bronze / silver / gold) with a pre-registered multi-phase promotion system: candidate strategies ideated by LLMs pass statistical screening, out-of-sample validation, and paper-trading stages, each with explicit pre-specified acceptance criteria. Phase-2 validation infrastructure operational.
- Structured rejection logging: every rejected strategy records the failed check, decision threshold, and realized statistic, producing a dataset of LLM-generated-artifact failure modes. Framework generalizes to AI-safety evaluation: the same verification loop applies to any domain where models propose high-stakes actions and need automated, auditable gating.

agentbreed · Lead Author

Mar 2026 – Present

Pre-registered empirical study of multi-agent LLM configuration optimization

- Pre-registered protocol (timestamped, git-pinned) before any real-model run. Benchmarks: ForecastBench, LiveCodeBench v6, GPQA Diamond. Models: Qwen3-32B-Instruct, Llama-3.3-70B-Instruct. 8-method optimization ladder (random search, coordinate descent, Bayesian opt, successive halving, genetic variants). Pre-specified analysis: linear mixed-effects with equivalence bounds (TOST) and Sobol-based sensitivity decomposition.
- Methodological transparency: publicly documented a variance-decomposition bug in a pilot (Sobol indices summed > 1.0, violating the pre-registered consistency check). Reran the pilot and published the deviation log before launching the main study.
- Test-gated evaluation harness: 509 unit and fuzz tests surfaced 23 issues before data collection, including a parse-failure bias (0.021 absolute score, 40% of the primary-hypothesis decision threshold). Per-finding audit log. Deployed on 2× H200 via vast.ai.

maze-rl-baselines · Lead Author

Feb 2026 – Apr 2026

Empirical study on reward-driven discovery failure in deep RL on procedural mazes

- Result: DQN, DRQN, PPO, and A2C with matched hyperparameter sweeps match a uniform random walk on 9×9 procedural mazes, while a behavior-cloning MLP of identical architecture, observation, and optimizer reaches 97.4% test success. A 5-line egocentric wall-following heuristic solves 100% of unseen 9×9 mazes; the best of seven HP-tuned modern reward-driven baselines (SB3 PPO, DQN, A2C across three learning rates; 70 total runs) reaches only 31.4%, statistically tied with Random at 32.7%.
- Headline finding: initializing DQN from the BC-distilled weights and fine-tuning with the same shaped reward collapses test success from 97.2% to 13.6% across all 5 seeds. Ablations consistent with optimization dynamics and distribution shift, not capacity or initialization.
- Replicated on 4 MiniGrid environments (DoorKey, FourRooms, MultiRoom, Unlock). Cover-time scaling on induced maze graphs matches the Alon–Benjamini–Lubetzky–Sodin non-backtracking prediction in finite settings. 3,500+ runs; ~40 GPU-hours plus H200 cluster time; public regenerable data.
- github.com/tejasnaladala/maze-rl-baselines

SEAL Lab, University of Washington · Research Associate

Mar 2025 – Nov 2025

PI: Prof. Alex Mamishev · Sensors, Energy, and Automation Laboratory

- PPG drowsiness-detection wearable: transimpedance front-end, 4th-order Butterworth (0.5–5 Hz), 12-bit ADC at 100 Hz, adaptive noise cancellation achieving 95% motion-artifact rejection.
- Piezoelectric hull-integrity sensor (U.S. Navy program): PZT array, charge amplifier, real-time 512-point FFT at 10 kHz, acoustic-emission classification separating impact from fatigue-crack propagation.
- Co-authored technical sections of 2 funded grant proposals across a 7-person team; surveyed 40+ papers on PPG signal processing and embedded anomaly detection.

NIIST-CSIR, India · Research Intern

Jun 2024 – Mar 2025

PI: Dr. Suraj Soman · Centre for Sustainable Energy Technologies (world-record DSC efficiency group)

- Fabricated and characterized dye-sensitized and perovskite solar cells (N719 on mesoporous TiO₂; I⁻/I₃⁻ electrolyte; spiro-OMeTAD over MAPbI₃).
- Characterized 20+ device architectures via IV curves (Dyename Toolbox), photo-induced absorption spectroscopy (PIA) for recombination kinetics, IPCE for spectral response, and XRD with Scherrer grain-size analysis.

INDUSTRY**PlasmaX · Founder & CTO**

May 2023 – Jan 2026

Industrial plasma-activated-water disinfection system for post-harvest produce

- Engineered the first industrial-scale continuous plasma-activated water disinfection system for post-harvest produce processing: chemical-free, 4× cheaper than chlorine sanitation, delivering 99.7% microbial reduction.
- Designed a custom 18 kV AC flyback driver at 23 kHz switching frequency with <2 W quiescent draw; implemented an STM32-based PID control loop managing gas flow rate, voltage regulation, and thermal management via MOSFET-driven power stages with real-time ADC feedback.
- Closed \$2M seed at \$8M post-money valuation; sold first production unit (\$180K) to CSIR-CFTRI India; contracted 10 revenue-generating system deployments across Indian commercial packhouses.

- Filed provisional patent on venturi-plasma integration architecture; co-authored 3 peer-reviewed publications characterizing RONS generation kinetics and antimicrobial efficacy of plasma-activated water.

PUBLICATIONS

Google Scholar: 35 total citations · h-index 3 · i10-index 2

1. Misra NN, Naladala T, Alzahrani KJ. Design of systems for plasma activated water (PAW) for agri-food applications. *Journal of Physics D: Applied Physics* 57(49), 493003 (2024). IOP. [12 citations] doi.org/10.1088/1361-6463/ad77de

2. Misra NN, Naladala T, Alzahrani KJ, Sreelakshmi VP, Negi PS. Design of a continuous plasma activated water (PAW) disinfection system for fresh produce industry. *Innovative Food Science & Emerging Technologies* 97, 103845 (2024). Elsevier. [9 citations] doi.org/10.1016/j.ifset.2024.103845

3. Misra NN, Sreelakshmi VP, Naladala T, Alzahrani KJ, Negi PS. Design and construction of a continuous industrial scale cold plasma equipment for fresh produce industry. *Innovative Food Science & Emerging Technologies* 97, 103840 (2024). Elsevier. [14 citations] doi.org/10.1016/j.ifset.2024.103840

Preprints in preparation:

- Connectome Architecture Benchmark: Parchment Labs (submitted to Nature Communications).
- Reward-driven discovery failure in deep RL on procedural mazes: maze-rl-baselines (preprint).
- Pre-registered empirical study of multi-agent LLM configuration optimization: agentbreed.

OPEN SOURCE PROJECTS & COMPETITIONS

OpenAI Parameter Golf (Model Craft Challenge)

Apr 2026

- Beat the prior 2nd-place submission on the track `_10min_16mb` track (10-minute training wallclock, 16 MB parameter budget, BPB objective). Trained a 3.7M-parameter decoder-only transformer (6 layers, 384 hidden dim, 6 heads) on 12M FineWeb-Edu documents with a custom BPE tokenizer pipeline.
- Ran a 31-candidate factorial sweep on 8x H100 SXM, varying tokenizer (BPE vocab size, merge count), architecture (width, depth, attention heads), optimizer (AdamW / Lion variants, LR schedule, warmup, weight decay), and training schedule (batch size, grad accumulation, data ordering). Watchdog terminated divergent runs early; pairwise config selection used Welch's t on BPB with wallclock-aware tiebreaking. Pluggable backend ran identically on free-GPU providers.

Forge · Open-source multi-agent LLM runtime

Feb 2026

- Provider-agnostic orchestration (4,400+ LOC, Python / TypeScript): declarative YAML agents, LiteLLM routing across 8 providers, pluggable memory (SQLite + ChromaDB), 6 built-in tools, multi-agent patterns (sequential / parallel / supervisor), FastAPI + WebSocket observability with a Next.js dashboard. 150+ clones in week one. github.com/tejasnaladala/Forge

Engram · Continuous-learning framework

Apr 2026

- Open-source continuous learning framework (Rust / TypeScript / Python, 300K+ LOC) implementing biologically-inspired neural regions with local Hebbian learning rules and neuromodulatory reward signals: no backprop, optimizer, or training loop. Prioritized experience replay with TD-error-weighted sampling; associative + episodic memory with persistent cross-session state; safety kernel that gates dangerous actions before execution. Online adaptation in real time on edge hardware without GPU retraining.

WireML · Node-graph workbench on foundation models

Apr 2026

- Teachable Machine re-imagined for modern foundation models. Browser-first via WebGPU with zero-backend hosted mode; Python runtime auto-detects CUDA / MPS / MLX / ROCm / DirectML / XPU / CPU. Every capability (data sources, backbones, training heads, evaluators, deployers) is a draggable, wireable node. CLIP, DINOv2, SigLIP, Whisper, and MediaPipe out of the box with Hugging Face Hub model support. Portable graphs serialize to single JSON files that run identically in-browser and on the local runtime.

Knowledge Engine

Apr 2026

- Platform for converting saved content into compounding technical leverage: multi-platform ingestion (bookmarks, highlights, articles, notes) into a queryable knowledge graph with semantic search, enabling reasoning across disparate sources.

claude-nexus · Multi-agent coordination for Claude Code

Mar 2026

- Real-time multi-agent coordination (TypeScript, ~3K LOC): WebSocket relay with auto peer discovery, shared context store with causal ordering, priority task queue with work-stealing, adversarial code-review protocol, sandboxed remote command execution, cross-network auto-tunneling.

ai-agent-city · Autonomous AI civilization simulator

Mar 2026

- Persistent agents, real economy with trading and resource management, emergent social dynamics, multi-agent interaction in a simulated urban environment; agents develop relationships, trade, form organizations, and evolve behaviors over time.

Mimic · Robotics imitation learning pipeline

Mar 2026

- End-to-end pipeline (41 modules, 103 tests): FastAPI + aiortc WebRTC streaming MuJoCo Franka Panda 7-DOF sim at 60 FPS, damped Jacobian pseudoinverse IK, LeRobot v3 recording (Parquet + MP4), dual-policy training (Action Chunking Transformer + DDPM Diffusion Policy with cosine noise schedule), ONNX export, Hugging Face Hub integration.

iCordion · iPhone → virtual accordion

Mar 2026

- Real-time accelerometer-to-bellows mapping with dynamic filter sweep; musette tuning via three detuned oscillators per note (Web Audio API); two octaves of chromatic treble keys plus six bass/chord buttons; self-signed HTTPS local server for iOS accelerometer access.

deluisnotthesolulu · Local-first relationship analytics

Feb 2026

- 100% local processing: no data leaves the device. On-device ML inference analyzes communication patterns, attachment styles, and relationship dynamics from chat screenshots with privacy as a hard constraint.

Cerulean Robotics · Open-source underwater robots for lake-health monitoring

Nov 2025

- Vectored 5-thruster config on Pixhawk 4 / ArduSub with MAVLink; RPi 5 H.264 video via GStreamer over CAT6 tether (<100 ms latency); custom KiCad MOSFET power distribution (4x IRF3205, 6S Li-ion 22.2 V / 5 Ah, current sensing + OCP); waterproofed acrylic housings rated to 100 m with dual O-ring seals. 28 units deployed across Washington; 4-figure MRR; King County LOI for Lake Sammamish environmental auditing.

tejas-os · Personal portfolio

Apr 2026

- Personal portfolio site ("Showing stuff that my resume falls too short for"). TypeScript. tejasnaladala.com

TECHNICAL SKILLS

ML & LLMs: PyTorch, Hugging Face (Transformers, Datasets, Accelerate, PEFT), transformer training (BPE, DDP / FSDP, bf16 mixed-precision), quantization (INT4/INT8, GPTQ, AWQ), LLM pre-training and fine-tuning (LoRA), multi-agent systems, adversarial debate architectures, RAG + vector search (ChromaDB, HNSW), LLM orchestration (OpenAI, Anthropic, vLLM, Ollama, LiteLLM multi-provider routing), Stable Baselines 3, Gymnasium, MiniGrid.

Research methodology: pre-registered study design, timestamped analysis plans, linear mixed-effects models, TOST equivalence testing, Sobol sensitivity decomposition, bootstrap CIs, Welch's t, fuzz and property-based testing, adversarial test harnesses, knowledge graphs, HNSW retrieval.

Languages & frameworks: Python (Expert), TypeScript / JavaScript (Expert), Rust (Advanced), C / C++ (Advanced), Java (Proficient), SQL (Proficient), MATLAB / Simulink (Capable), Next.js / React (Expert), FastAPI / Node.js (Advanced), WebRTC / WebSocket (Advanced), ONNX (Advanced).

Robotics & simulation: MuJoCo, imitation learning (ACT, Diffusion Policy), inverse kinematics (Jacobian), MAVLink / ArduSub, GStreamer, ROS / Gazebo.

Embedded & hardware: circuit design (analog/digital), PCB design (KiCad), STM32 / Arduino / RPi, Pixhawk / ArduSub, MOSFET power electronics, high-voltage PSU design (flyback, 18 kV+), real-time DSP (FFT, Butterworth, PID), LTspice, oscilloscope / logic analyzer, SMD + PTH soldering.

Design & fabrication: Fusion 360 / SolidWorks / Rhino 8, 3D printing (FDM / SLA), CNC / VMC / lathe, laser / waterjet cutting, injection molding.

Infrastructure: SLURM / HPC (H100, H200 via vast.ai and Jarvis Labs, A40), Docker, Linux, Git, AWS (EC2 / Lambda), Vercel, FastAPI / WebSocket, SQLite, ChromaDB, Jupyter.

AWARDS & RECOGNITION

- Y Combinator AI Startup School 2026: SF cohort
- OpenAI Parameter Golf (Model Craft Challenge): beat prior 2nd-place submission on track_10min_16mb (2026)
- Carnegie Mellon Venture Challenge: 2nd Place out of 150+ teams (\$4,500) (2026)
- UW Science & Technology Showcase: Grand Prize (Pack Ventures) + Best Pitch Award (\$2,750) (2026)
- Red Bull Basement: 2nd Place
- Buerk Center Prototype Grant (\$2,000)
- Lavin Entrepreneurship Fellow: four-year selective program at UW
- Provisional patent: venturi-plasma integration architecture (PlasmaX)

ADDITIONAL

Languages: English, Hindi, Telugu, Mandarin.

Interests: Powerlifting (6 days/week PPL), beatboxing, public speaking, entrepreneurial mentoring.